

Review

Global studies on trust in science suggest new theoretical and methodological directions

Jan Pfänder^{1,a}, Niels G. Mede² and Viktoria Cologna¹

Public trust in science is vital for tackling global challenges. Recently, global surveys and Many Labs collaborations have begun to broaden the scope of research. However, these studies have also highlighted theoretical and methodological challenges. Here, we review these challenges and argue that beyond expanding geographical coverage, greater conceptual clarity and harmonized measures are essential to improve comparability across studies on trust in science. We conclude by encouraging reflection on the normative assumptions that currently guide research on trust in science.

Addresses¹ Department of Environmental Social Sciences, Swiss Federal Institute of Aquatic Science and Technology (Eawag), Switzerland² Strategic Communication Group, Department of Social Sciences, Wageningen University & Research, the NetherlandsCorresponding author: Pfänder, Jan (jan.pfaender@eawag.ch)^a Department of Environmental Social Sciences, Swiss Federal Institute of Aquatic Science and Technology (Eawag), Überlandstrasse 133, Switzerland CH-8600 Dübendorf

We need public trust in science to effectively address pressing challenges of humanity: Crises such as climate change are so complex in nature that evaluating their risks and mitigation options requires extensive expertise. People who report trusting science are more supportive of such expertise. For example, they are more likely to accept human made climate change [1], engage in pro-environmental behavior [2–4], and get vaccinated ([5], for Covid-19 in particular, see Refs. [6,7]).

Much research on public trust in science has focused on the United States (US) and Europe. However, tackling global issues such as climate change or pandemics requires trust in science at a global scale. Many Labs

studies present a promising approach to address this gap by pooling resources and expertise from research teams worldwide, allowing for more feasible and harmonized data collection.

The Trust in Science and Science-related Populism (TISP) Many Labs study with almost 72,000 participants from 68 countries has illustrated the benefits of this approach [8]: First, it has revealed that trust dynamics can be highly context dependent. For example, conservative and right-leaning individuals trust scientists less in Europe and in the US, but in several countries in Africa and Southeast Asia the opposite is the case [8]. In most surveyed countries, there was no significant relationship between political orientation and trust. Second, it has established cross-culturally robust findings. For example, contrary to a popular narrative [9,10], the TISP study [8] found no signs of a crisis of public trust in science: In each of the 68 surveyed countries, trust was above the scale midpoint (mean = 3.62; sd = 0.70; Scale: 1 = very low, 2 = somewhat low, 3 = neither high nor low, 4 = somewhat high, 5 = very high). Third, the TISP study [8] has produced new normative perspectives on the role of scientists in society: Beyond being mere knowledge producers, in most countries, a majority of people want scientists to engage in policymaking.

Despite these advances, global approaches also raise new questions due to contrasting findings. For example, the Wellcome Global Monitor [11] reported that greater income inequality was negatively associated with trust in scientists, while the TISP study [8] found the opposite. Numerous differences in research design—from measuring trust to country selection—make it difficult to draw meaningful conclusions from these divergent findings.

This example illustrates that broadening the geographical scale is not enough to robustly advance knowledge on trust in science. As the study of trust in science is growing, there is a need to discuss theoretical and methodological approaches and how they can be improved. In this review, we outline challenges and discuss normative considerations that we hope will inform future research.

Current Opinion in Psychology 2026, **67**:102215

This review comes from a themed issue on Trust in Science and Beyond (2026)

Edited by Jonathan Lewis-Jong and Natalia J Zarzeczna

For complete overview about the section, refer [Trust in Science and Beyond \(2026\)](#)

Available online 14 November 2025

<https://doi.org/10.1016/j.copsyc.2025.102215>2352-250X/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Theoretical challenges

When investigating public trust in science, researchers first need to ask: How should we conceptualize trust? Clearly defining referents of trust will likely improve comparability across studies and produce more robust results.

Trust in what?

The term ‘science’ evokes different representations among different individuals or groups [12]. Science may be understood as a body of literature, an institution, a method, or individual scientists. These representations matter for trust. For example, in the US, people tend to trust scientific methods more than scientific institutions—particularly among less-educated segments of the population [13].

Trust in science is also domain-specific: people do not trust all scientific disciplines equally. For instance, people around the world perceive climate scientists as less trustworthy than scientists in general [14]. In the US [15–18], but also in France [19], disciplines such as biology or physics enjoy considerably more trust than economics or sociology. Scientific knowledge is another example of domain-specific trust. In the US, essentially everyone tends to overwhelmingly trust basic scientific knowledge [20]. Yet, on certain contentious and politicized topics, small but concerning minorities reject the scientific consensus [21]—on climate change, see, e.g., Refs. [22,23], and on GMOs, see, e.g., Ref. [24]. Domain-specific trust can also relate to different character traits of scientists, such as competence, integrity, or benevolence [25]. Research from the US has shown that people tend to perceive scientists as competent but not particularly warm [26].

Being precise about the construct under study and its domain-specific aspects can result in more meaningful insights. For example, the TISP study [8] assessed the perceived trustworthiness of scientists with a multi-item index based on a large systematic review of trust dimensions [27]. This more detailed scale allowed for fine-grained insights: While the public perceived scientists’ competence to be high, perceptions of integrity and openness were notably weaker. These findings point to a key challenge for science communication: not only demonstrating expertise but also integrity.

To identify and validate theoretical dimensions of trust, qualitative studies play a crucial role. For example, an analysis of open-ended answers to the question “What makes a scientist trustworthy?” in Ireland [28] validated the dimensions of trustworthiness used in the TISP study [8].

Perceptions of trustworthiness vs. acts of trusting

Most research on public trust in science has investigated perceptions of trustworthiness, while much less

attention has been paid to specific acts of trusting [29,30]. Perceptions of trustworthiness are often conceptualized as evaluations of scientists’ character traits [25]. Behavioral trust, by contrast, reflects people’s willingness to make themselves vulnerable to the behaviors or decisions of scientists [29]. For instance Ref. [27], developed a scale to capture how much influence people think scientists should have on public policy and their personal life. However, conceptualizations of behavioral trust differ across studies [30], and it is not always clear how they can be distinguished from beliefs. For example, does believing in human-made climate change already constitute an act of trust in climate scientists?

In the context of social trust, declared trust has been shown to be associated with behavioral trust [31]. This also seems to be the case for science: During the Covid-19 pandemic, declared trust in science was one of the most important correlates of protection intentions (e.g., social distancing [32,33]) and supporting protective policies [e.g., implementing curfews [6]. However, experimentally manipulating perceptions of trustworthiness, researchers could not demonstrate that they *caused* protection intentions against Covid-19 [34]. As this example suggests, it is often not clear how perceptions of trustworthiness translate into specific acts of trusting.

Methodological challenges

Ambiguities at the theoretical level are frequently mirrored in the varying approaches researchers use for sample selection and measurement. Such differences have led to inconsistent findings, hindering cumulative knowledge generation.

Sample selection

Selecting study participants involves several methodological choices—such as whether to conduct fieldwork online or by telephone, use probability-based or quota sampling, and rely on weighting or stratification procedures—all of which can influence findings on trust in science [35]. Global evidence suggests that one aspect of sample selection is particularly important: country selection.

In the US, there is a strong partisan divide regarding trust in science: Democrats tend to express more trust in science than Republicans [36–38]. Based on this evidence, it would be reasonable to assume that liberal political orientation is an important predictor of higher trust. However, global analyses reveal a more nuanced picture: While in Europe and in the US more right (vs. left) leaning and more conservative (vs. liberal) individuals tend to trust scientists less, the opposite is true in several countries in Africa and Southeast Asia [8]. Moreover, the role of political orientation can shift across domains of science: For instance, conservatism is associated with greater climate change skepticism in

North America and Europe, but not in other countries [39,40].

A similar pattern emerges for religiosity. Studies have found that in the US [38] or in the Netherlands [41], more religious individuals express lower trust in scientists. The TISP study [8] revealed important nuances: It showed that religiosity is, on average, slightly positively associated with trust in scientists, driven mostly by a positive association in several Muslim-majority countries. This aligns with global findings on religion and attitudes towards science more generally [42], and with qualitative research suggesting that Muslims tend to view teachings of their religion and science as overlapping [43].

These examples illustrate how relying exclusively on samples from the Global North, where survey research on trust in science is institutionalized and well accessible, bears the risk of overgeneralization. Achieving more diverse samples remains a major challenge. Even in the TISP study [8], several regions such as Central Asia, the Middle East, and Oceania were underrepresented.

Measurement

Using different measures to assess the same theoretical construct can lead to inconsistent results. This concerns both trust in science and the covariates assessed in relation to trust.

Some survey projects assess trust in science with very general questions, often relying on single items.^b Because people can hold different representations of science, and because trust is domain-specific, it remains unclear what exactly these questions capture [27,29]. As a consequence, general single-item measures of trust in science may produce inconsistent findings across populations and hinder the comparability of findings across studies.

Global studies on public perceptions of science have often relied on aggregate measures of trust. For example, the Wellcome Global Monitor surveys use a five-item index to assess trust in scientists [11,45].^c Aggregate measures help address concerns about single-item

questions. What they cannot address, however, is the issue of comparability: if studies use different trust indices and arrive at different conclusions, it is difficult to reconcile the discrepancies [46].

The issue of measurement heterogeneity also extends to common covariates of trust. On the one hand, new operationalizations can strengthen confidence in established findings. For instance, research has found that the link between scientific knowledge and attitudes toward science is weak, and largely absent for issue-specific attitudes [47]. Using different measures, the TISP study [8] confirmed this result: Instead of measuring science knowledge with narrow factual quizzes, they used average national scores from the Program for International Student Assessment (PISA) and found no statistically significant association between countries' average trust in scientists and their average PISA scores.

On the other hand, measuring covariates differently can also challenge previously established findings. For example, many global studies have identified education [48] and science education [11] as two of the strongest correlates of trust in science. By contrast, the TISP study [8] tested this relationship using only a coarse distinction between tertiary versus all other forms of education (including none), finding only a weak positive association.

Another example is religion: The TISP study [8] found a positive correlation between religiosity and trust in science in several muslim-majority countries. By contrast, another global study [48] used data on religious affiliation not at the country, but at the individual level, and found that Muslims held the least favorable views of science among several religious groups.^d Differences in model specification and country selection, further impede comparability.^e

As research on trust in science increasingly relies on more culturally diverse samples, our understanding of regional patterns of trust has improved. However, greater conceptual clarity and convergence on validated, theoretically grounded measures is needed for findings to be comparable across contexts and theory to advance. Collaborative efforts, such as Many Labs studies, provide promising pathways to achieve this.

Normative considerations

Approaches to studying trust in science are not value-agnostic: they are guided, implicitly or explicitly, by normative assumptions about what public trust in

^b For example, the US General Social Survey has tracked trust in science for 50 years, but relies on a single item: "Would you say you have a great deal of confidence, only some confidence, or hardly any confidence at all in the scientific community?". Similarly, the Pew research center relies on a single-item question, asking people about how much confidence they have that "scientists act in the best interests of the public" [44].

^c These items are: "How much do you trust scientists in this country? A lot, some, not much or none at all?"; "In general, how much do you trust scientists to find out accurate information about the world? A lot, some, not much or none at all?"; "How much do you trust scientists working in colleges/universities in this country to do their work with the intention of benefiting the public? A lot, some, not much or none at all?"; "How much do you trust scientists working in colleges/universities in this country to be open and honest about who is paying for their work? A lot, some, not much or none at all?"; "How much do you trust scientists working for companies in this country to do their work with the intention of benefiting the public? A lot, some, not much or none at all?"

^d The other religious groups were Catholic, Protestant, Orthodox, Jewish, Buddhist and Hindu.

^e [48] use several additional covariates in their model, and do not include majority-Muslim countries, such as Malaysia, Egypt, Morocco, and Bangladesh, where the TISP study [8] found a positive association between religiosity and trust.

science should look like. Moving forward, we believe the field needs to reflect more explicitly on its underlying normative assumptions.

One important example is the long-standing influence of deficit thinking. For decades, research on public perceptions of science has emphasized public “deficits”, ranging from a lack of knowledge to a lack of trust [49,50]. This deficit thinking has three major consequences: First, it takes trust in science for granted and treats it as a rational default. Theoretical approaches in psychology [51–53], communication [54,55], political science [56], and sociology [12,57] have focused on why certain groups deviate from that default. But more research is needed on why people across the globe tend to trust science in the first place [8], in spite of knowing little about it [58]. Addressing this gap is essential if we want to understand how trust can be fostered and sustained [59]. Second, deficit thinking suggests that the more people trust science, the better. For topics with a well-established scientific consensus—e.g., the existence of human made climate change, or the safety of vaccines—more trust in science can indeed be expected to lead to better societal outcomes. But public trust in science should not be blind, either: for disruptive knowledge and emerging technologies to be in the public’s best interest, the public needs to critically evaluate them [60]. A research agenda on trust should therefore not only ask how to increase trust, but also when healthy skepticism is warranted. Third, deficit thinking tends to stigmatize the public as ignorant. This pathologization of trust deficits does not acknowledge that low trust in science may have legitimate reasons, such as past failures and scientific misconduct [61].

To meaningfully advance research on trust in science, three key aspects deserve greater consideration: theoretical clarity about what trust is, methodological rigor to measure it consistently across contexts, and normative deliberation of what levels and forms of trust are desirable. Elevated levels of public trust in science should not encourage complacency. As demonstrated by recent political developments, science is under attack even in countries such as the US, where most people trust science at least to some extent, but where trust is increasingly polarized along political ideologies and where narratives of a loss of public trust are instrumentalized to justify budget cuts for US universities [62]. In the face of these challenges, advancing research on public trust in science is perhaps more important than ever.

Credit author statement

Author roles were classified using the Contributor Role Taxonomy (CRediT; <https://credit.niso.org/>) as follows: *Jan Pfänder*: conceptualization and writing — original draft. *Niels G. Mede*: conceptualization and writing —

review & editing. *Viktoria Cologna*: conceptualization, writing — review & editing, and supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

References of particular interest have been highlighted as:

* of special interest

** of outstanding interest

1. Bogert JM, Buczny Harvey, Ellers J: **The effect of trust in science and media use on public belief in anthropogenic climate change: a meta-analysis.** *Environ Commun* 2024, **18**:484–509, <https://doi.org/10.1080/17524032.2023.2280749>.
2. Cologna V, Siegrist M: **The role of trust for climate change mitigation and adaptation behaviour: a meta-analysis.** *J Environ Psychol* 2020, **69**, 101428, <https://doi.org/10.1016/j.jenvp.2020.101428>.
3. Hornsey MJ, Harris EA, Bain PG, Fielding KS: **Meta-analyses of the determinants and outcomes of belief in climate change.** *Nat Clim Change* 2016, **6**:622–626, <https://doi.org/10.1038/nclimate2943>.
4. Todorova B, Steyrl D, Hornsey MJ, Pearson S, Brick C, Lange F, **et al.:** **Machine learning identifies key individual and nation-level factors predicting climate-relevant beliefs and behaviors.** *Npj Climate Action* 2025, **4**:46, <https://doi.org/10.1038/s44168-025-00251-4>.
5. Sturgis P, Brunton-Smith I, Jackson J: **Trust in science, social consensus and vaccine confidence.** *Nat Hum Behav* 2021, **5**: 1528–1534, <https://doi.org/10.1038/s41562-021-01115-7>.
6. Algan Y, Cohen D, Davoine E, Foucault M, Stantcheva S: **Trust in scientists in times of pandemic: panel evidence from 12 countries.** *Proc Natl Acad Sci* 2021, **118**, e2108576118, <https://doi.org/10.1073/pnas.2108576118>.
7. Lindholt MF, Jørgensen F, Bor A, Petersen MB: **Public acceptance of COVID-19 vaccines: cross-national evidence on levels and individual-level predictors using observational data.** *BMJ Open* 2021, **11**, e048172, <https://doi.org/10.1136/bmjopen-2020-048172>.
8. Cologna V, Mede NG, Berger S, Besley J, Brick C, Joubert M, **et al.:** **Trust in scientists and their role in society across 68 countries.** *Nat Hum Behav* 2025:1–18, <https://doi.org/10.1038/s41562-024-02090-5>.
9. Nichols TM: *The death of expertise: the campaign against established knowledge and why it matters.* New York, NY: Oxford University press; 2017.
10. Jamieson KH: **Crisis or self-correction: rethinking media narratives about the well-being of science.** *Proc Natl Acad Sci* 2018, **115**:2620–2627, <https://doi.org/10.1073/pnas.1708276114>.
11. Wellcome Global Monitor: *Wellcome global monitor 2018.* 2018.

12. Gauchat G: **The cultural authority of science: public trust and acceptance of organized science.** *Publ Understand Sci* 2011, 20:751–770, <https://doi.org/10.1177/0963662510365246>.
 13. Achterberg P, De Koster W, Van Der Waal J: **A science confidence gap: education, trust in scientific methods, and trust in scientific institutions in the United States, 2014.** *Publ Understand Sci* 2017, 26:704–720, <https://doi.org/10.1177/0963662515617367>.
 14. Ghasemi O, Cologna V, Mede NG, Stanley SK, Strahm N, Ross R, et al.: **Gaps in public trust between scientists and climate scientists: a 68 country study.** *Environ Res Lett* 2025, 20, 061002, <https://doi.org/10.1088/1748-9326/add1f9>.
 15. Altenmüller MS, Wingen T, Schulte A: **Explaining polarized trust in scientists: a political stereotype-approach.** *Sci Commun* 2024, 46:92–115, <https://doi.org/10.1177/10755470231221770>.
- People across the political spectrum hold stereotypes about scientists' political orientation (e.g., "scientists are liberal"). These stereotypes matter for trust in science: Both in the US and in Germany, conservatives tend to distrust scientists more when they believe that scientists are liberal.
16. Gligorić V, Kleef GA van, Rutjens BT: **How social evaluations shape trust in 45 types of scientists.** *PLoS One* 2024, 19, e0299621, <https://doi.org/10.1371/journal.pone.0299621>.
- Trust in scientists is relatively high, on average, but varies substantially between different disciplines. For example, among the most trusted scientists were meteorologists and physiologists, among the least trusted were pharmacologists and political scientists.
17. Gauchat G, Andrews KT: **The cultural-cognitive mapping of scientific professions.** *Am Sociol Rev* 2018, 83:567–595, <https://doi.org/10.1177/0003122418773353>.
 18. Gligorić V, Van Kleef GA, Rutjens BT: **Political ideology and trust in scientists in the USA.** *Nat Hum Behav* 2025, <https://doi.org/10.1038/s41562-025-02147-z>.
 19. Pfänder J, Mercier H: *The French trust more the sciences they perceive as precise and consensual.* 2025, https://doi.org/10.31219/osf.io/k9m6e_v1.
 20. Pfänder J, Kerzreho L, Mercier H: **Quasi-universal acceptance of basic science in the United States.** *Publ Understand Sci* 2025, <https://doi.org/10.1177/09636625251364407>.
- In four studies, they asked US Americans—including flat Earthers, climate change deniers and vaccine skeptics—whether they accepted basic scientific facts (e.g., electrons are smaller than atoms). Acceptance of the scientific consensus was very high in the sample as a whole (95.1%), but also in every sub-sample (e.g., among people who say they do not trust science at all: 87.3%; among participants who completely endorsed flat Earth theory: 87.2%).
21. **Pew Research Center: Major gaps between the public, scientists on key issues.** 2015.
 22. Stockemer D, Bordeleau J-N: **Understanding climate change conspiracy beliefs: a comparative outlook.** *Harvard Kennedy School Misinformation Review* 2024, 5, <https://doi.org/10.37016/mr-2020-162>.
 23. Uscinski J, Enders A, Klofstad C, Seelig M, Drochon H, Premaratne K, et al.: **Have beliefs in conspiracy theories increased over time?** *PLoS One* 2022, 17, e0270429, <https://doi.org/10.1371/journal.pone.0270429>.
 24. Schröder TB: **Don't tell me what I don't want to hear! politicization and ideological conflict explain why citizens have lower trust in climate scientists and economists than in other natural scientists.** *Polit Psychol* 2023, 44:961–981, <https://doi.org/10.1111/pops.12866>.
 25. Hendriks F, Kienhues D, Bromme R: **Measuring laypeople's trust in experts in a digital age: the Muenster Epistemic Trustworthiness Inventory (METI).** *PLoS One* 2015, 10, e0139309, <https://doi.org/10.1371/journal.pone.0139309>.
 26. Fiske ST, Dupree C: **Gaining trust as well as respect in communicating to motivated audiences about science topics.** *Proc Natl Acad Sci* 2014, 111:13593–13597, <https://doi.org/10.1073/pnas.1317505111>.
 27. Besley JC, Lee NM, Pressgrove G: **Reassessing the variables used to measure public perceptions of scientists.** *Sci Commun* 2021, 43:3–32, <https://doi.org/10.1177/1075547020949547>.
 28. Roche J, Hurley M, Fowler K, McConville A, Taylor A, Mede NG, et al.: **Science and society in Ireland: examining public trust in scientists against a global background.** *Ir Educ Stud* 2025: 1–22, <https://doi.org/10.1080/03323315.2025.2488804>.
 29. Besley JC, Tiffany LA: **What are you assessing when you measure "trust" in scientists with a direct measure?** *Publ Understand Sci* 2023, 32:709–726, <https://doi.org/10.1177/09636625231161302>.
 30. Besley JC, Benitez Gonzalez A, Tiffany LA: **Differentiating behavioral trust and trustworthiness beliefs to improve science communication practice and research.** *Curr Opin Psychol* 2026, 67, 102192, <https://doi.org/10.1016/j.copsyc.2025.102192>.
 31. Ahmed AM, Salas O: **The relationship between behavioral and attitudinal trust: a cross-cultural study.** *Rev Soc Econ* 2009, 67:457–482, <https://doi.org/10.1080/00346760902908625>.
 32. Dohle S, Wingen T, Schreiber M: **Acceptance and adoption of protective measures during the COVID-19 pandemic: the role of trust in politics and trust in science.** *Soc Psychol Bulletin* 2020, 15, e4315, <https://doi.org/10.32872/spb.4315>.
 33. Pagliaro S, Sacchi S, Pacilli MG, Brambilla M, Lionetti F, Bettache K, et al.: **Trust predicts COVID-19 prescribed and discretionary behavioral intentions in 23 countries.** *PLoS One* 2021, 16, e0248334, <https://doi.org/10.1371/journal.pone.0248334>.
 34. Wingen T, Posten A-C, Dohle S: *When correlation does not imply causation no evidence for causal effects of trust in science on pandemic-related protection intentions.* 2025, https://doi.org/10.31219/osf.io/th953_v1.
- In studies conducted during the COVID-19 pandemic (March 2021–February 2022), they experimentally manipulated trust in science by presenting participants with real-world statements from scientists—some of which at the point of the study had turned out to be primarily correct (high trust condition), some of which incorrect (low trust condition). They did not find that this manipulation increased protection intentions (e.g. physical distancing, mask wearing) among participants.
35. Smith BK, Jensen EA: **Critical review of the United Kingdom's "gold standard" survey of public attitudes to science.** *Publ Understand Sci* 2016, 25:154–170, <https://doi.org/10.1177/0963662515623248>.
 36. Krause NM, Brossard D, Scheufele DA, Xenos MA, Franke K: **Trendsamericans' trust in science and scientists.** *Public Opin Q* 2019, 83:817–836, <https://doi.org/10.1093/poq/nfz041>.
 37. Druckman JN, Schulman J, Safarpour AC, Baum M, Ognyanova K, Kenny M, et al.: *Continuity and change in trust in scientists in the United States: demographic stability and partisan polarization.* 2024, <https://doi.org/10.2139/ssrn.4929030>.
- Trust in science in the U.S. is remarkably stable, and so are its demographic correlates: women, Black, rural, religious, non-college educated, and lower/working class individuals exhibit less trust. The study suggests that the partisan gap (Democrats trusting more, Republicans less) can be explained by low-trust demographic strata shifting partisan allegiances. This suggests that political ideology might be a less important factor of polarization than demographic inequities.
38. Azevedo F, Jost JT: **The ideological basis of antiscientific attitudes: effects of authoritarianism, conservatism, religiosity, social dominance, and system justification.** *Group Process Intergr Relat* 2021, 24:518–549, <https://doi.org/10.1177/1368430221990104>.
 39. Rutjens BT, Sengupta N, Der Lee RV, Van Koningsbruggen GM, Martens JP, Rabelo A, et al.: **Science skepticism across 24 countries.** *Soc Psychol Personal Sci* 2022, 13:102–117, <https://doi.org/10.1177/19485506211001329>.
 40. Hornsey MJ, Harris EA, Fielding KS: **Relationships among conspiratorial beliefs, conservatism and climate scepticism**

- across nations. *Nat Clim Change* 2018, **8**:614–620, <https://doi.org/10.1038/s41558-018-0157-2>.
41. Rutjens BT, Van Der Lee R: **Spiritual skepticism? Heterogeneous science skepticism in the Netherlands.** *Publ Understand Sci* 2020, **29**:335–352, <https://doi.org/10.1177/0963662520908534>.
 42. McPhetres J, Jong J, Zuckerman M: **Religious Americans have less positive attitudes toward science, but this does not extend to other cultures.** *Soc Psychol Personal Sci* 2021, **12**: 528–536, <https://doi.org/10.1177/1948550620923239>.
 43. Thigpen CL, Johnson C, Funk C: *On the intersection of science and religion.* 2020.
 44. Kennedy AT, Brian: *Public trust in scientists and views on their role in policymaking.* 2024.
 45. Wellcome Global Monitor: *Wellcome global monitor 2020: covid-19.* 2020.
 46. Mede NG, Cologna V, Berger S, Besley J, Brick C, Joubert M, et al.: **Public Communication about Science in 68 Countries: Global Evidence on How People Encounter and Engage with Information about Science.** *Science Communication* 2025. <https://doi.org/10.1177/10755470251376615>.
 47. Allum N, Sturgis P, Tabourazi D, Brunton-Smith I: **Science knowledge and attitudes across cultures: a meta-analysis.** *Publ Understand Sci* 2008, **17**:35–54, <https://doi.org/10.1177/0963662506070159>.
 48. Noy S, O'Brien TL: **Science for good? The effects of education and national context on perceptions of science.** *Publ Understand Sci* 2019, **28**:897–916, <https://doi.org/10.1177/0963662519863575>.
 49. Bauer MW, Allum N, Miller S: **What can we learn from 25 years of PUS survey research? Liberating and expanding the agenda.** *Publ Understand Sci* 2007, **16**:79–95, <https://doi.org/10.1177/0963662506071287>.
 50. Scheufele DA: **Thirty years of sciencesociety interfaces: what's next?** *Publ Understand Sci* 2022, **31**:297–304, <https://doi.org/10.1177/09636625221075947>.
 51. Hornsey MJ, Fielding KS: **Attitude roots and Jiu Jitsu persuasion: understanding and overcoming the motivated rejection of science.** *Am Psychol* 2017, **72**:459–473, <https://doi.org/10.1037/a0040437>.
 52. Rutjens BT, Večkalov B: **Conspiracy beliefs and science rejection.** *Curr Opin Psychol* 2022, **46**, 101392, <https://doi.org/10.1016/j.copsyc.2022.101392>.
 53. Lewandowsky S, Oberauer K: **Worldview-motivated rejection of science and the norms of science.** *Cognition* 2021, **215**, 104820, <https://doi.org/10.1016/j.cognition.2021.104820>.
 54. Mede NG, Schäfer MS: **Science-related populism: conceptualizing populist demands toward science.** *Publ Understand Sci* 2020, **29**:473–491, <https://doi.org/10.1177/0963662520924259>.
 55. Mede NG, Schäfer MS, Metag J, Klinger K: **Who supports science-related populism? A nationally representative survey on the prevalence and explanatory factors of populist attitudes toward science in Switzerland.** *PLoS One* 2022, **17**, e0271204, <https://doi.org/10.1371/journal.pone.0271204>.
 56. Druckman JN: **Threats to science: politicization, misinformation, and inequalities.** *Ann Am Acad Polit Soc Sci* 2022, **700**:8–24, <https://doi.org/10.1177/00027162221095431>.
 57. Gauchat G: **The legitimacy of science.** *Annu Rev Sociol* 2023, **49**:263–279, <https://doi.org/10.1146/annurev-soc-030320-035037>.
- Argues that politicization of science in the US needs to be seen in its broader cultural context. Science has played a crucial role in legitimizing the modern regulatory state. Consequently, conservative distrust of science reflects deeper structural tensions with the rational–legal authority of modern governance.
58. National Academies of Sciences, Engineering, and Medicine: *Science literacy: concepts, contexts, and consequences.* Washington, D.C.: National Academies Press; 2016, <https://doi.org/10.17226/23595>.
 59. Pfänder J: *A cognitive perspective on trust in science.* 2025. PhD thesis.
 60. Scheufele DA, Krause NM, Freiling I, Brossard D: **What we know about effective public engagement on CRISPR and beyond.** *Proc Natl Acad Sci* 2021, **118**, e2004835117, <https://doi.org/10.1073/pnas.2004835117>.
 61. Krause N, Scheufele D, Freiling I, Brossard D: **The trust fallacy.** *Am Sci* 2021, **109**:226, <https://doi.org/10.1511/2021.109.4.226>.
 62. Mervis J: **Trump's science adviser defends funding cuts as a chance to 'revitalize' u.s. science | science | AAAS.** *Science* 2025.
- ## Further information on references of particular interest
4. Among 19 candidate variables, trust in climate science was the ** most important predictor of climate change belief and policy support.
 8. The TISP Many Labs assessed the perceived trustworthiness of ** climate scientists with a 12-item scale, covering the dimensions of competence, benevolence, integrity and openness. They found moderately high levels of trust in scientists across the globe.
 15. People across the political spectrum hold stereotypes about sci- ** entists' political orientation (e.g., "scientists are liberal"). These stereotypes matter for trust in science: Both in the US and in Germany, conservatives tend to distrust scientists more when they believe that scientists are liberal.
 16. Trust in scientists is relatively high, on average, but varies sub- * stantially between different disciplines. For example, among the most trusted scientists were meteorologists and physiologists, among the least trusted were pharmacologists and political scientists.
 20. In four studies, they asked US Americans—including flat Earth- * ers, climate change deniers and vaccine skeptics—whether they accepted basic scientific facts (e.g., electrons are smaller than atoms). Acceptance of the scientific consensus was very high in the sample as a whole (95.1%), but also in every sub-sample (e.g., among people who say they do not trust science at all: 87.3%; among participants who completely endorsed flat Earth theory: 87.2%).
 34. In studies conducted during the COVID-19 pandemic (March * 2021–February 2022), they experimentally manipulated trust in science by presenting participants with real-world statements from scientists—some of which at the point of the study had turned out to be primarily correct (high trust condition), some of which incorrect (low trust condition). They did not find that this manipulation increased protection intentions (e.g. physical distancing, mask wearing) among participants.
 37. Trust in science in the U.S. is remarkably stable, and so are its * demographic correlates: women, Black, rural, religious, non-college educated, and lower/working class individuals exhibit less trust. The study suggests that the partisan gap (Democrats trusting more, Republicans less) can be explained by low-trust demographic strata shifting partisan allegiances. This suggests that political ideology might be a less important factor of polarization than demographic inequities.
 57. Argues that politicization of science in the US needs to be seen in its broader cultural context. Science has played a crucial role in legitimizing the modern regulatory state. Consequently, conservative distrust of science reflects deeper structural tensions with the rational–legal authority of modern governance.